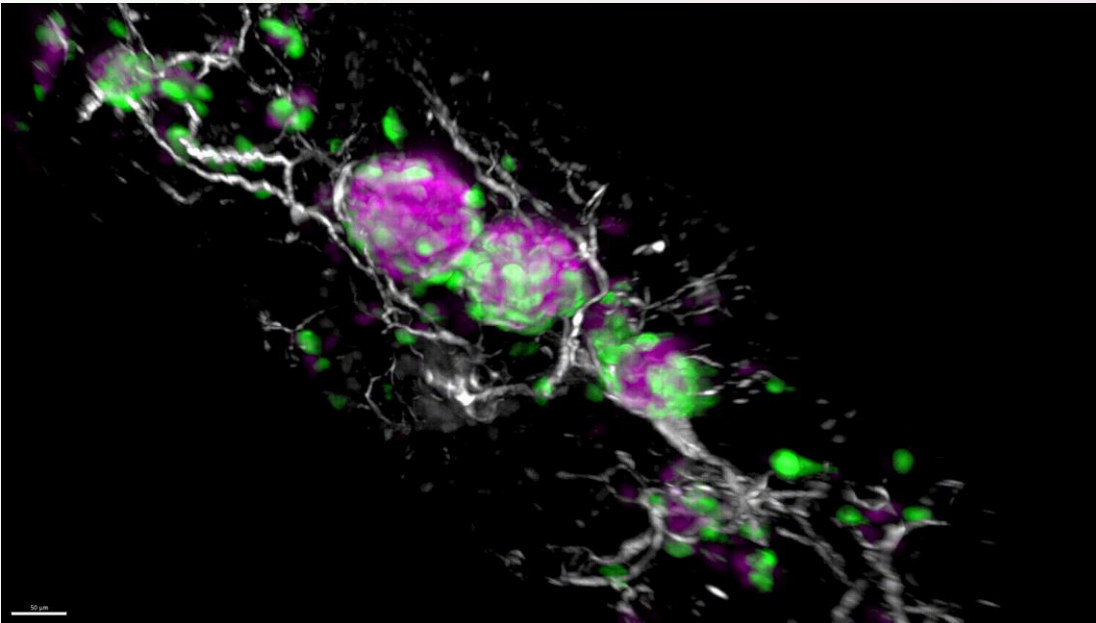


Regulatory networks of crucial cell identity genes in embryonal development could unveil potential drug targets in many pathologies like diabetes

- Understudied role of glucagon (Gcg) producing alpha cells in diabetes
- Glucagon fluctuations contribute to diabetes progression
- Alpha cells are important for paracrine signalling to insulin producing beta cells
- How are these cells generated during embryonal development?
- Aim: Potential targets for restoration of their function



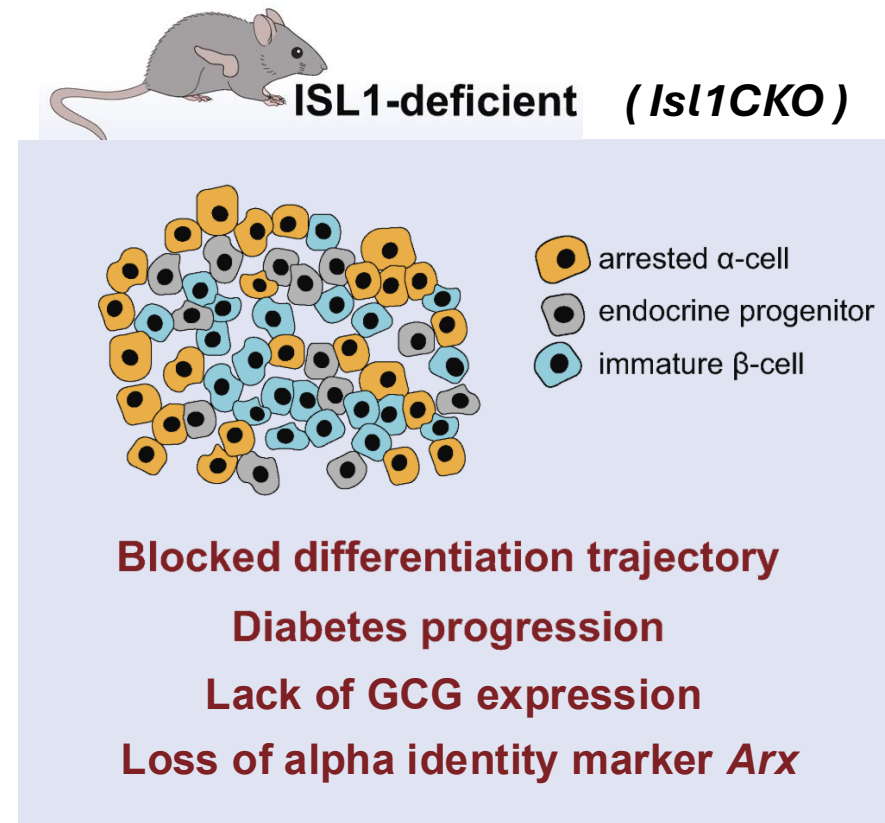
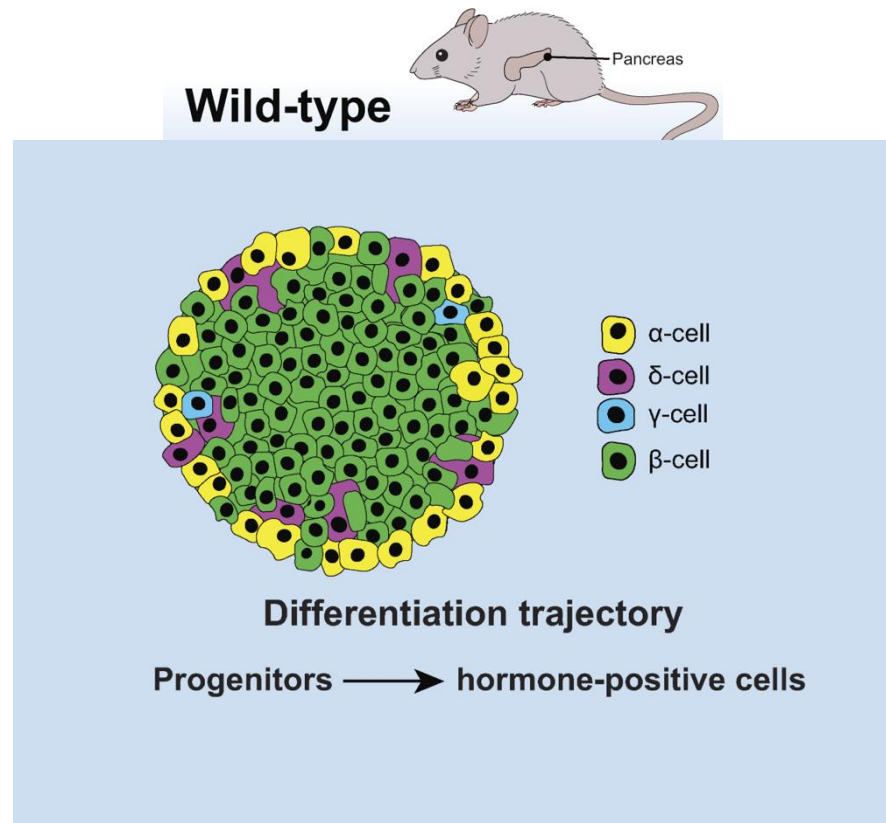
GCG



ISL1-deficient mice struggle with differentiation of pancreatic endocrine lineages, which leads to diabetes



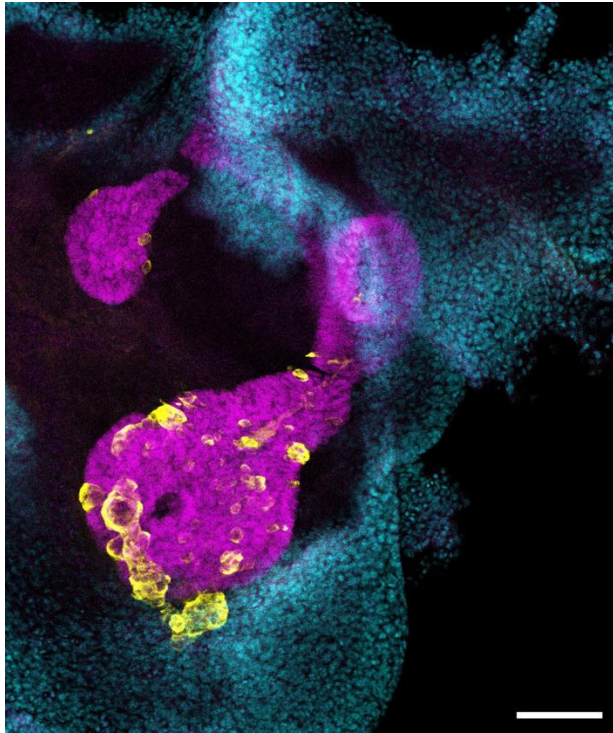
Dr. Gabriela Pavlínková
Head of laboratory Molecular Pathogenetics
Czech Academy of Sciences
Prague



Loss of glucagon in secondary transition in *Isl1CKO* islets

Primary transition

E9.5 - E12.5

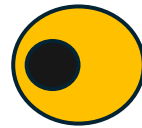


GCG PDX1

ISL1CKO



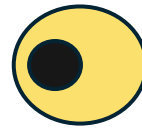
E12.5



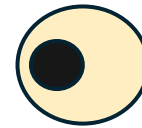
Arx
Gcg



E14.5



E18.5

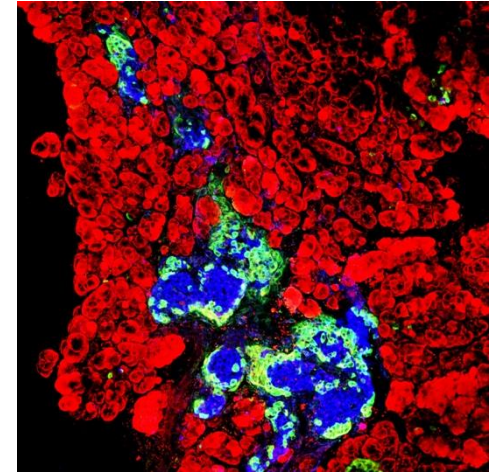


Arx
Gcg

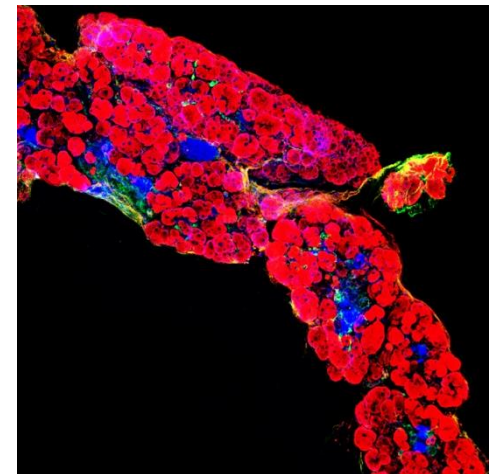
Secondary transition

E13.5 - E18.5

control

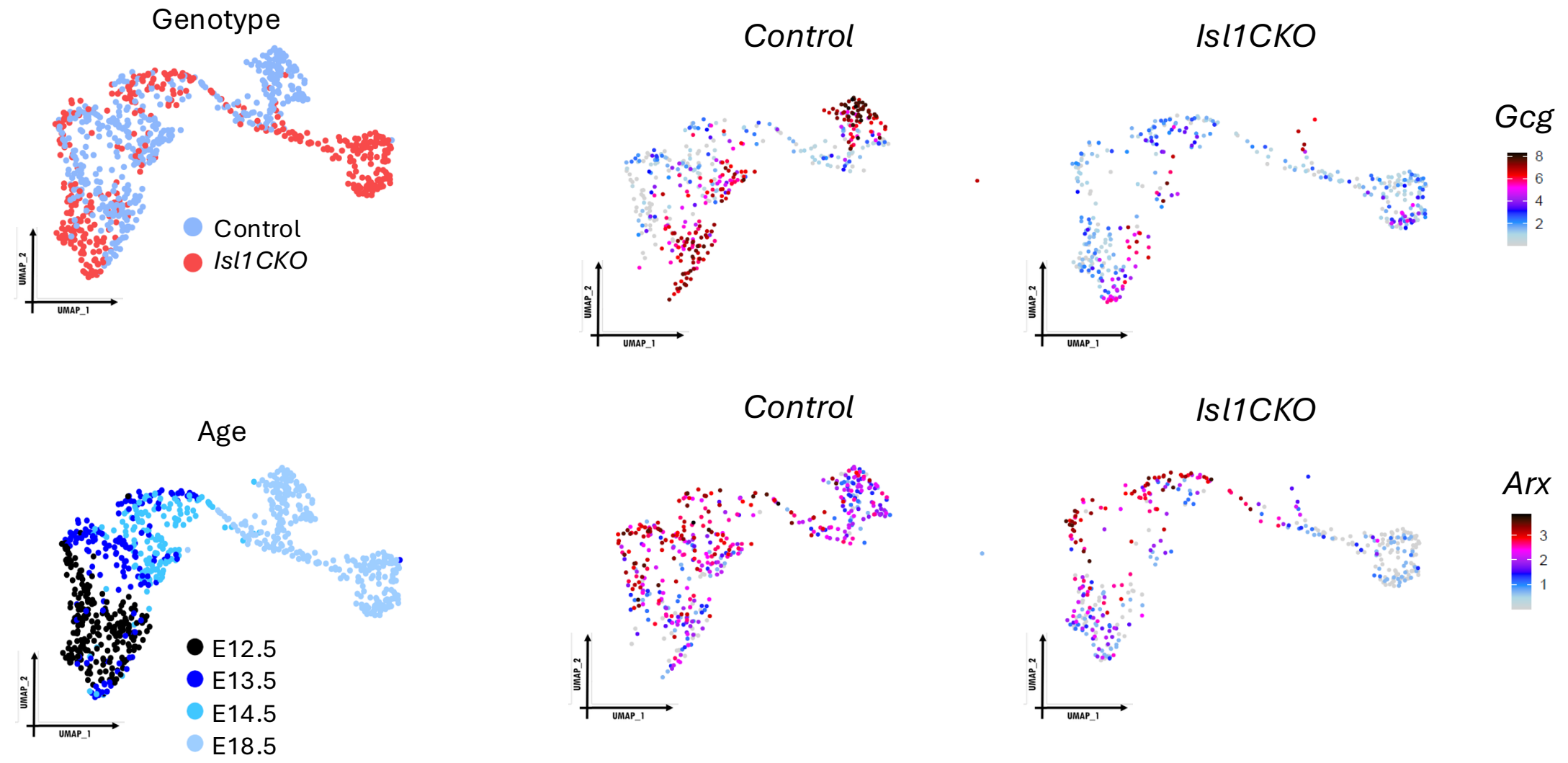


Isl1CKO



GCG AMY INS

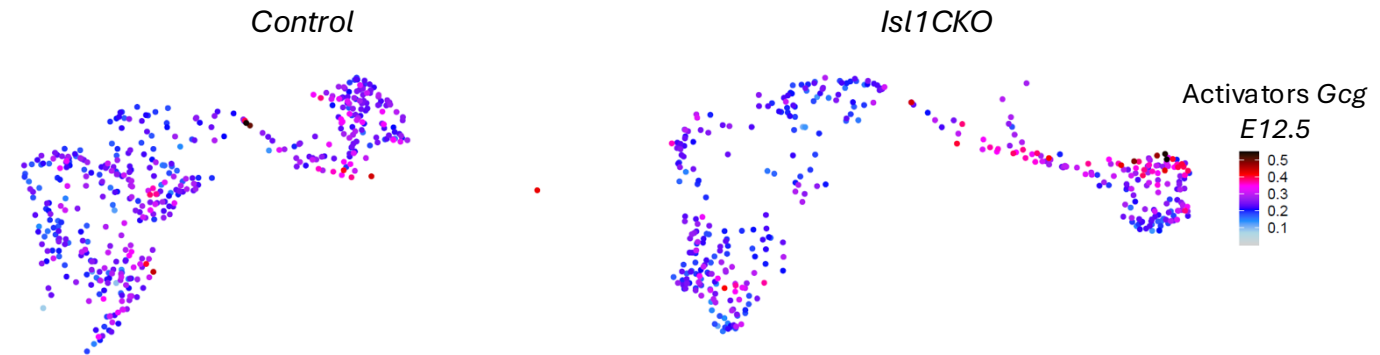
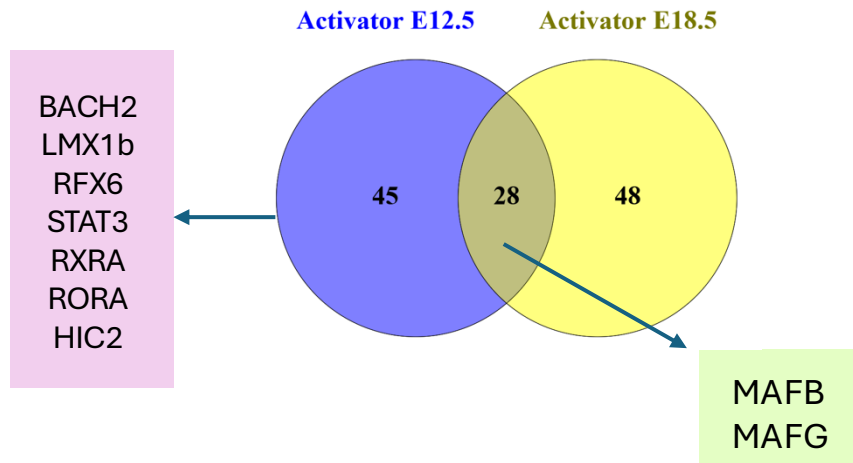
Single cell transcriptomics of embryonal alpha cells shows loss of *Gcg* and *Arx* transcription after secondary transition (E18.5) in *Isl1CKO* cells



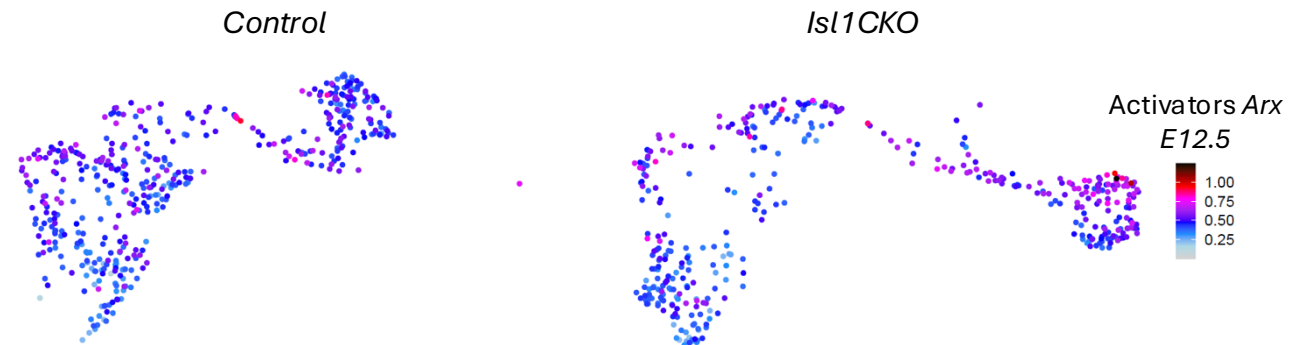
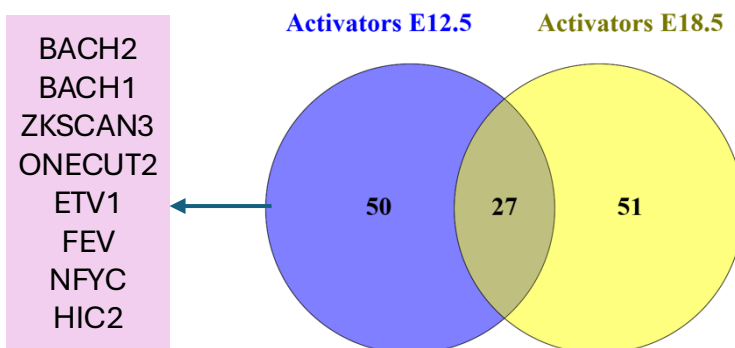
Primary transition – how is Glucagon and *Arx* expression maintained?

NetRem analysis

Unique activators and common activators of *Gcg* gene transcription



Unique activators and common activators of *Arx* gene transcription

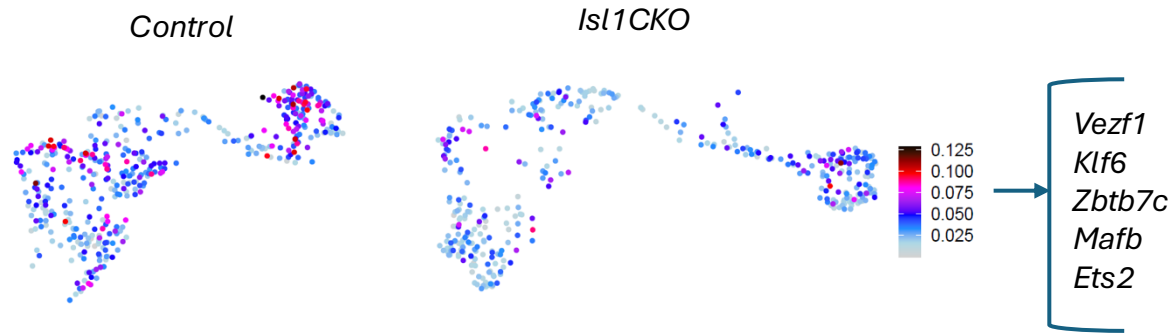


Secondary transition – how is Glucagon and *Arx* expression lost?

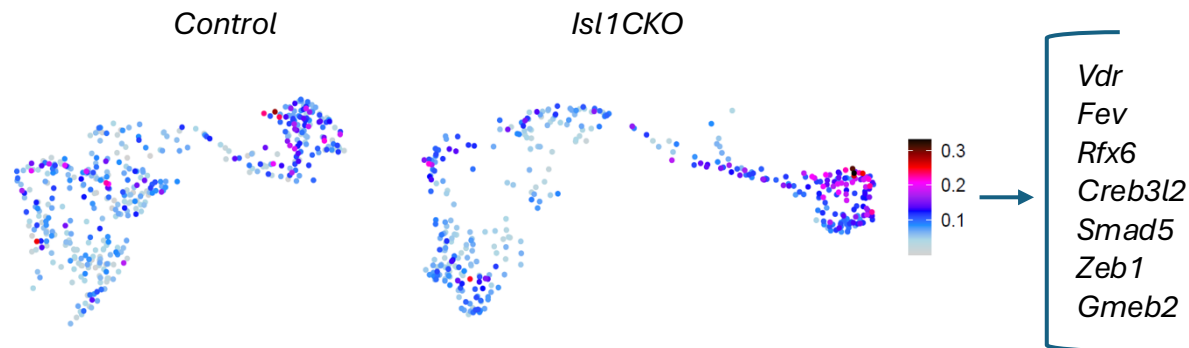
NetRem analysis

Gcg

Downregulated activators at E18.5

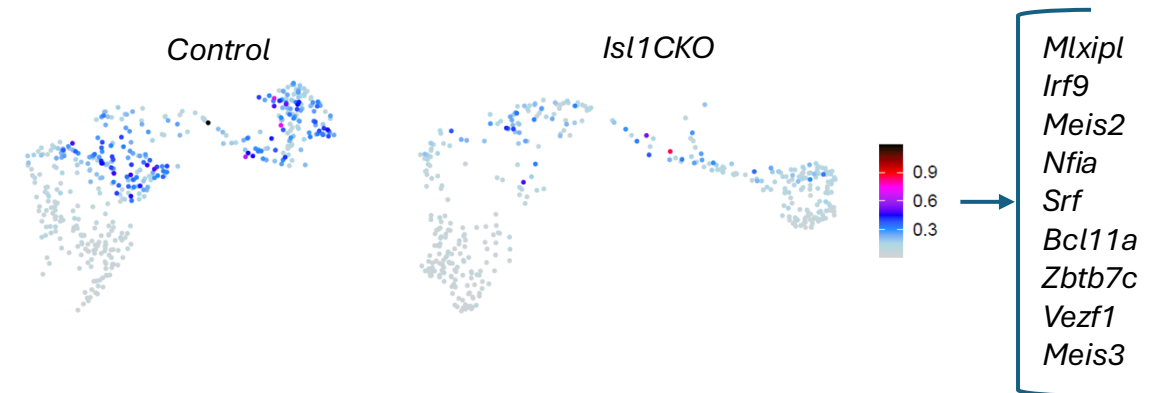


Upregulated repressors at E18.5

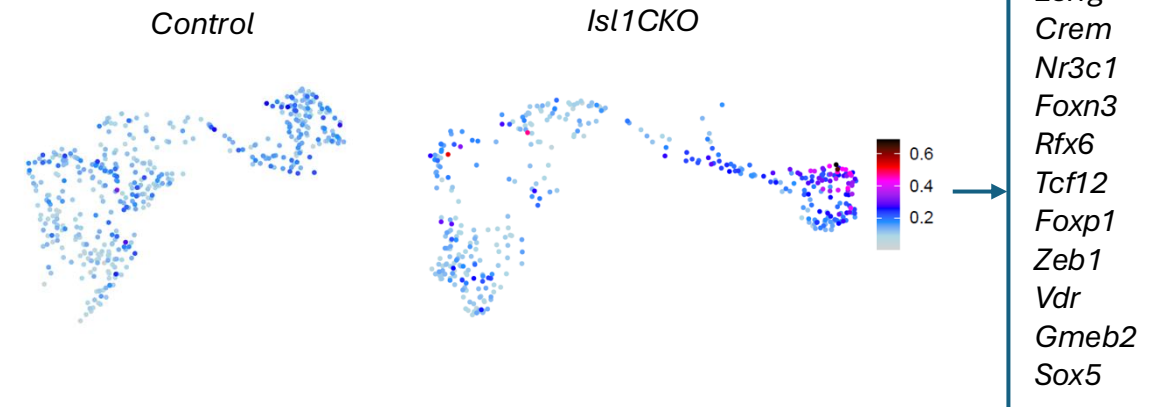


Arx

Downregulated activators at E18.5



Upregulated repressors at E18.5

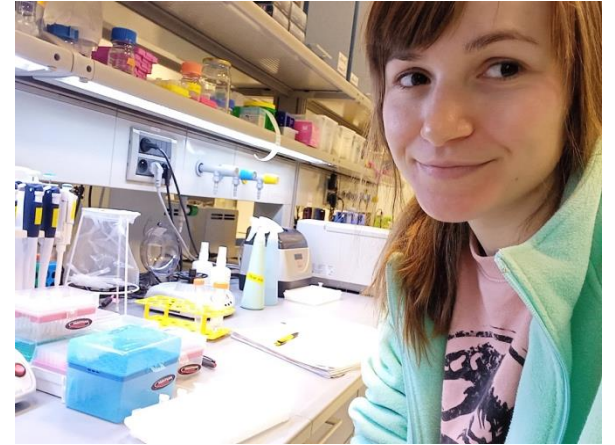


New insights from NETREM:

- ARX function is dependent on its expression level
- Higher ARX levels – repression of *Gcg* expression, while being necessary for α -lineage induction
- Lower ARX levels – maintenance of *Gcg* transcription
- *Gcg* and *Arx* regulation is age-dependent, with unique activators and repressors during the primary and secondary transitions of pancreatic development

“ NetREm helped us discover potential regulators of alpha identity marker in *in vivo* conditions, and regulators of glucagon expression during early and late embryonal development, which could help find drug targets for improvement of alpha cell function in diabetes. ”

Valeria, PhD candidate



“ It’s a great prediction tool for discovering relationships between regulatory proteins and building their networks in cell type and stage specific manner. ”

Věra, Postdoc



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ISL1 Restricts Progenitor Programs and Promotes β -Cell Maturation, Revealing Sex Differences in Diabetes Progression

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Efficient control of enterochromaffin versus islet differentiation from human pluripotent stem cell-derived pancreatic progenitors

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Check for updates

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RESEARCH

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ISL1 controls pancreatic alpha cell fate and beta cell maturation

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